

Dmitry Messerman

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Location: Haifa, Israel (until April, 2027), Herzliya (from April 2027)

Professional Summary

Ph. D. in Electrical Engineering

M.Sc. in Electrophysics

EDA CAD tool expert with over 20 years of multi-disciplinary experience in the development, support, and deployment of Computer Aided Design and Silicon Debug tools for high-speed and analog custom VLSI design.

Renowned for multi-disciplinary expertise, problem solving and algorithm development.

Work Experience

Principal Engineer at Intel Corporation, Haifa, Israel 1992 - 2025

- Led the development, deployment customization and support of in-house and commercial Electronic Design Automation tools in multiple areas
 - Translated user and system requirements into working solutions
 - Led cross-team engineering efforts
 - Worked directly with internal stakeholders and leading EDA vendors
- Led development of parasitic networks synthesis from VLSI design layout
 - Developed heuristics for accurate modeling of resistivity parasitic effects in VLSI design with $O(N)$ complexity – has been used in production for ~10 years, patent granted
- Led development of Reliability Validation tool for VLSI design – has been used in production for ~20 years
- Developed novice algorithm for fast inductance extraction from the design layout – patent granted
- Deployed novice multi-grid algorithm for CPU power grid (about 10^8 elements) simulation with $O(N)$ complexity
- Developed surrogate model for silicon heat dissipation in tester environment based on diffusion equations solver
 - Found design issue in the CPU tester, which led to DUT (Device Under Test) overheating
- Led development of silicon debug tools
 - Platform HW/SW co-simulation for BIOS Memory Reference Code validation
 - Structure-Based Functional Tests (SBFT) coverage improvement based on genetic algorithm.
 - Found a major flaw in CPU architecture
- Recognized for exceptional debugging and problem-solving capabilities in CAD software and Linux

Physicist and software engineer in FrantzTech Ltd 1990 - 1992

- Co-developed ultrasound image processing algorithm for early detection of diffused liver diseases

Researcher in Research Institute for High-Voltage Direct Current Power Transmission 1982-1990

- Modeling and simulation of large-scale power transmission systems, including transient phenomena, multi-conductor line behavior, corona discharged, lightning surge and environmental effect
 - Analysis of the environmental impact of electric fields from extra-high-voltage power lines
 - Development of fundamental algorithms for calculating the modal parameters of multi-conductor power transmission lines, taking into account the characteristics of non-homogeneous soil.
 - Modeling of the distortion and attenuation of lightning surge waves in long-span power transmission lines

Education

Ph. D. in Electrical Engineering, St. Petersburg Polytechnic University, Russian Federation

M.Sc. in Electrophysics, St. Petersburg Polytechnic University, Russian Federation

Skills

- Programming Languages: Python, C/C++, MATLAB, Perl, Tcl
- Development environment: GitHub
- Languages: English, Hebrew, Russian

Publications and patents

1. Yan J., Aydiner A., **Messerman D.**, Zhu J., Camacho Mora A., Liao J., Ma M., Jiao D. Efficient Equalization Optimization Algorithm for Signaling Analysis. Nonlinear System 2019 IEEE MTT-S International Conference on Numerical Electromagnetic and Multiphysics Modeling and Optimization (NEMO).
2. Reisfeld G., **Messerman D.**, Bone N., Lazar A. VARAN: variability analysis for memory cell robustness. Proc. SPIE 6925, 69250L, 2008.
3. **Messerman D.**, Seltser M. Reorganizing rectangular layout structures for improved extraction. - United States Patent 7389001 B2, 2008.
4. Chakravarty S., Ben-Noon Y., Chiprout E., Mazumder M., **Messerman D.** Inductance Modeling. - United States Patent 7325208 B2, 2008.
5. **Messerman D.** Mathematical modeling of wave processes in high-voltage overhead power lines across a wide range of frequencies and soil electrical parameters. - Ph. D. Thesis, Moscow, 1989.

Recommendations

Ronen Cohen, Senior Staff Software Engineer

I had the privilege of managing Dmitry for several years, during which he provided exceptional support for various Analog tools and flows for teams in Israel. Dmitry is an outstanding CAD Engineer whom I strongly recommend. His unique blend of deep technical expertise and professionalism is evident in every tool and workflow he develops or maintains. He has a remarkable ability to translate complex requirements into practical, straightforward solutions, while clients greatly value his clear communication and can-do attitude. No matter how challenging the task, Dmitry was always the person I could rely on to deliver results.

Gideon Reisfeld, Ph.D. in Physics

Dmitry was my manager from 2006 to 2011. The technical depth and contributions I experienced under his leadership were truly exceptional. His expertise and innovative thinking in electrical engineering, mathematics, and computer science brought a transformative leap in the quality of the CAD solutions we developed.

Among his many achievements were evaluations in Static Timing Analysis within Intel's timing tool, the development of a thermal back-annotation tool for debugging tester thermal performance, and robust memory analysis techniques.

In short, if I were to choose someone as the ultimate debugger and a true technology maverick, Dmitry would be my first choice.